



Picas y Fijas

Strategy guide

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-  **Fija (F):** right symbol, right position.
-  **Pica (P):** right symbol, another position.
-  **Nothing:** that symbol is not in the code.

The central idea. A good try is not the one that looks close to the code: it is the one that leaves the fewest possible codes, whatever the answer turns out to be.

This guide does not give opinions: it measures. The openings and second moves it recommends were computed with the game's own engine, the same one that grades every try in "How it played out" when a game ends. It then explains how the plan changes with a clock and with limited tries, which rules earn more ranking points and how to choose your own code.

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Four digits without repeats, played by the engine

5.27
tries on average

61%
in five tries or fewer

7
tries at most

These are the rules of the code of the day. Chapters 2 and 3 explain how to get close to those figures.

1 The method

Five steps that work for every variant, with numbers or with colors.

1	Explore. Open with a code that tests many symbols at once. Chapter 2 says which one, variant by variant.
2	Record. Write down every try with its fijas and its picas separately. F+P tells you how many symbols you hit; the fijas, how many are already in place.
3	Filter. Rule out every code that, if it were the secret, would not have given exactly the same clues on all your tries.
4	Split. While many candidates remain, play the try that spreads them best across different answers. A probing try does not have to be able to win.
5	Solve. When one or very few remain, play a candidate; if you can choose, the one that best separates the others.

How to read the answer

Clue	What it means	What to do
	None of those symbols is in the code.	Rule them all out. It is one of the most informative answers.
	They are in the code, but none in that position.	Move them next time you use them.
	Two positions are already right; you do not know which.	Keep one or two where they are and change the rest: the next clue will tell you which they were.
	F+P below the number of positions: symbols are missing.	Bring in new symbols without throwing away what you know about the ones you hit.
	F+P equal to the number of positions: you have every symbol.	Bring in no more: all that is left is ordering (chapter 4).

Mix early, isolate late. In the first moves it pays to do two things at once: move what you already tried and bring in new symbols. Once you know which symbols they are, change one thing per turn, so you know what produced the clue.

2 The opening, measured

Before the first try you know nothing, so the only thing that matters about an opening is its shape: how many different symbols it has and how many it repeats. The game's engine computes how many codes remain on average after each shape. The fewer, the better.

No repeats: every opening is worth the same

Any code with different symbols leaves exactly the same. There is no secret opening to memorize: always open with the same one — so the answers become familiar — and spend your thinking on the second move.

Rules	Codes	Opening	Left on average
3 positions	720	0 1 2	180 (25.0%)
4 positions	5,040	0 1 2 3	931 (18.5%)
5 positions	30,240	0 1 2 3 4	4,747 (15.7%)
6 positions	151,200	0 1 2 3 4 5	22,163 (14.7%)

With repeats: it depends on how many symbols there are

Here some shapes are better than others. The table shows what share of the codes remains, on average, after opening with all different symbols, with one pair or with two.

Rules	Codes	All different	One pair	Two pairs
Numbers · 3 positions	1,000	012 26.2% · optimal	001 34.8% · solid	—
Numbers · 4 positions	10,000	0123 18.4% · optimal	0012 20.3% · solid	0011 26.1% · solid
Numbers · 5 positions	100,000	01234 14.6% · optimal	00123 14.9% · optimal	00112 15.9% · optimal
Numbers · 6 positions	1,000,000	012345 12.4% · optimal	001234 12.2% · optimal	001123 12.2% · optimal
4 colors · 4 positions	256	ABCD 14.1% · solid	AABC 12.0% · optimal	AABB 12.3% · optimal
6 colors · 4 positions	1,296	ABCD 14.5% · optimal	AABC 14.3% · optimal	AABB 15.8% · solid
8 colors · 4 positions	4,096	ABCD 16.2% · optimal	AABC 17.2% · optimal	AABB 20.8% · solid

Share of the codes still possible after the opening, on average. "Optimal" and "solid" are the grades the game gives.

- **With numbers, open without repeats** up to four positions. Ten digits are a lot: a pair in the opening wastes coverage.
- **From five positions on** a pair no longer costs anything, and with six it is even the best choice.
- **With few colors it is the other way round:** with 4 and with 6 colors the pair comes in on the first try. With 8, all different rules again.

It is the same yardstick the game uses for its grades. Opening a four-digit code with repeats with 0011 leaves 42% more codes than 0123: in “How it played out” it is graded “Solid”, not “Optimal”. And playing the same digit in every position (0000) as an opening leaves 52% of the codes: “Wasted”.

3 The second move

Four digits, no repeats — the rules of the code of the day — after opening with 0123. For every answer, the try that leaves the fewest codes on average, searched among all codes and not only among those that can still be the secret.

Answer	Happens	Left	Best move	How it is built	Leaves on average
● ○ ○ ○	28.6%	1,440	1 0 4 5	2 moved + 2 new	254 / 295
● ● ○ ○	25.0%	1,260	1 4 3 5	2 moved + 2 new	202 / 322
● ● ○ ○	14.3%	720	0 1 4 5	2 kept in place + 2 new	108 / 184
● ○ ○ ○	9.5%	480	0 4 5 6	1 kept in place + 3 new	80 / 98
○ ○ ○ ○	7.1%	360	4 5 6 7	4 new	65 / 65
● ● ● ○	5.2%	264	1 4 3 5	2 moved + 2 new	45 / 103
● ● ● ○	4.3%	216	0 2 4 5	1 kept in place + 1 moved + 2 new	40 / 84
● ● ○ ○	3.6%	180	0 2 4 5	1 kept in place + 1 moved + 2 new	27 / 46
● ● ● ○	1.4%	72	0 2 4 5	1 kept in place + 1 moved + 2 new	12 / 28

“Leaves on average”: in bold, the codes left after the best move; next to it, those left by bringing in all-new digits (4567).

The rule that comes out of the table

- **No match at all:** all-new digits. It is the only case where that is best, and it only happens 7% of the time.
- **Only picas:** reuse two of the digits you played, in another position, and bring in two new ones.
- **Some fija:** keep one or two digits where they were — that is how you find out which were the fijas — and bring in new ones for the rest.
- **Do not follow a list.** After an answer of two picas, bringing in four new digits leaves 322 codes on average; the mixed move, 202.

What playing well looks like. Always choosing the try that leaves the fewest codes, the engine cracks a four-digit code without repeats in 5.27 tries on average: in five or fewer 61% of the time, and never in more than 7. The 8 tries of the code of the day are more than enough if you waste none.

With three digits

The same, after opening with 012. A three-digit code does not fall by itself either: the engine needs 5.07 tries on average and never more than 6.

Answer	Happens	Left	Best move	How it is built	Leaves on average
  	35.0%	252	1 0 3	2 moved + 1 new	59 / 68
  	29.2%	210	3 4 5	3 new	43 / 43
  	17.5%	126	0 3 4	1 kept in place + 2 new	29 / 34
  	8.8%	63	1 2 3	2 moved + 1 new	13 / 27

“Leaves on average”: in bold, the codes left after the best move; next to it, those left by bringing in all-new digits (345).

About 123 – 456 – 789. Covering the digits in blocks is easy to remember and it works, but it is only the best choice when the first block gave nothing. As soon as the opening gives a pica, reusing two digits elsewhere and bringing in one new digit teaches more than the next block; and as soon as F+P adds up to three you have all three digits: do not play the remaining block.

4 Ordering what you already have

When F+P equals the number of positions there is nothing left to look for: only placing. Without repeats, this is the plan by code length.

Positions	Opening	Plan
3	0 1 2	Mix reused and new digits until F+P adds up to 3, then stop bringing in new ones. If you open in blocks (012, 345, 678), remember the 9: it is the one left out.
4	0 1 2 3	The table in chapter 3. If F+P = 4, only the order is missing; if it is lower, replace two digits and move the ones you keep.
5	0 1 2 3 4	With a high F+P, keep the block and try controlled position changes. With a low F+P, swap two or three digits for 567 and measure how many come in.
6	0 1 2 3 4 5	The opening carries six of the ten digits, so F+P never drops below 2. Replace them two at a time with 67 and then 89; once the six digits are identified, order them by swaps.

The swap technique

If you know the symbols but not the order, start from one arrangement and swap just two positions. If the fijas go up by two, both are now right; if they go down by two, both were right before; if they go up or down by one, one of the two is — or was — in its place. Keeping everything else still is what makes the answer readable.

An example game

Three digits, no repeats, played by the engine against the code 250. “Left” counts the codes still possible after each clue.

Turn	Move and clue	Left	What it teaches
1	012 	63	Two of the digits 0, 1 and 2 are in the code, both out of place.
2	123 	18	It reuses 1 and 2 in another position and brings in 3. Only one scores: 3 is out and only one of 1 and 2 is in, so 0 is certain.
3	240 	5	2 and 0 in new positions, plus a new digit. Two fijas: of the 18 codes left, only those starting with 2 and ending in 0 fit. The middle digit is 5, 6, 7, 8 or 9.
4	567 	2	A probing move: 567 cannot be the code, but it spreads the 5 candidates over three answers. The pica says the middle digit is 5 or 7.
5	250 	–	250 and 270 were left. It was 250.

5 With repeats and with colors

With repeats you have to find out two things: which symbols appear and how many times each.

- **Each copy counts only once.** If you play 0011 and the code has a single 0, only one of your two zeros scores.
- **F+P does not count different symbols.** An F+P of 2 can be two symbols or the same symbol twice.
- **The pair, once few symbols are left.** With six or fewer still to test, a pair is already as good as all different; with four, it is better.
- **The quantity test.** Playing the same symbol in every position (1111) tells you exactly how many times it appears: as many as the fijas it gets. It is clear and it is expensive: keep it for the end, when the remaining candidates differ only in quantities.

A plan for four digits with repeats

- Open with 0123.
- If it gives nothing, six digits and 1,296 codes remain: it is like playing with six colors. Continue with a pair and two more digits: 4456 leaves 185 codes on average; 4567, 188.
- If $F+P = 4$, it is those four digits, once each: only the order is missing.
- If $F+P$ is between 1 and 3, the missing ones can be new digits or repeats of the ones you already have. Reuse two and bring in two, as without repeats, and try a pair as soon as you know which digits are still in play.

The typical mistake is to keep thinking of codes without repeats when the numbers no longer add up. If you have tested all ten digits in tries without repeats and their $F+P$ do not add up to the number of positions, some digit is repeated.

With colors

The logic is the same; what changes is the size of the palette. With few colors, repeats weigh more.

Palette	No repeats	With repeats, 4 positions
4 colors	With 4 positions the code uses all four colors: from the first try you are only ordering. With 3, one is missing, and F+P tells you whether it is the one you left out.	Open with a pair: AABC or AABB, almost the same.
6 colors	Open with different colors. With 4 positions, F+P tells you how many of the two colors you did not play are in the code.	AABC or ABCD, almost the same. AABB, the classic Mastermind opening, bounds the worst case better (256 codes against 276), but leaves more on average.
8 colors	It is like numbers: explore in blocks and stop as soon as F+P completes the positions.	All different (ABCD). The pair waits until few colors are left to test.

6 The clock and limited tries

The rules do not change the logic, but they do change how much exploring you can afford.

Rules	What changes	What to do
No limit, no clock	Nothing rushes you: you can play one try now and the next one hours later.	Go for certainty. There is room for probing tries that cannot win.
10 tries	Comfortable: with four digits and no repeats, the engine never needs more than 7.	Two or three exploring moves and, after that, only codes that can be the secret.
6 tries	Tight: with four digits and no repeats, the engine needs the sixth try 37% of the time, and a seventh 1.7%.	Not a single wasted move. From the fourth on, play only codes compatible with every clue.
Per-turn clock	At zero you lose that try and the turn passes to your opponent. With limited tries, that try counts too.	Always have a compatible code ready before looking for a better one.
Time bank	Only your clock runs while it is your turn. There is no pause, and whoever runs out of bank loses the game, whatever the position.	Play the opening and the second move from memory, and save the time for the end. With an increment, every quick move wins it back.
Correspondence	One or three days per move. The deadline runs even if you close the game; if it expires, you lose that try.	Time to spare: go over every clue, use the notebook and never play a code that contradicts one of them.

The last try

Whoever cracks the code first wins, but if the player who opened the round cracks it, the opponent gets one try to tie. If that try is yours, do not explore: play the candidate most likely to be the code. A draw is worth 4 points plus half the difficulty; a loss, 1.

Quick decision tree

After a move	Next decision
F+P = positions	Bring in no new symbols: work on the order and, with repeats, on the quantities.
F+P = 0	All those symbols are out. Bring in the missing ones.
In between	Reuse part of the move — in other positions if there were no fijas — and bring in new symbols.
2 to 5 candidates left	Play the code that gives a different answer for each of them, even if it is not one of them.
1 candidate left	Play it.

Common mistakes

- Playing a code that contradicts an earlier clue.
- Repeating a move that no longer tells the remaining candidates apart.
- Forgetting the 0 in numbers.
- Changing too many positions when you already have several fijas.
- Reading F+P as different symbols when repeats are allowed.
- Following a fixed sequence when a clue has already located every symbol.

7 What pays: the points

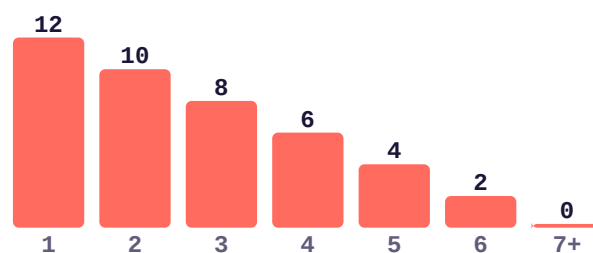
The ranking sorts by points, not by wins, and every month starts from zero. Only duels against another person hand out points, when they end.

$$\begin{array}{c} \mathbf{10} \\ \text{for the win} \end{array} + \begin{array}{c} \mathbf{0-14} \\ \text{difficulty} \end{array} + \begin{array}{c} \mathbf{0-12} \\ \text{economy} \end{array}$$

How the game ends	Points
You win by cracking the code	10 + difficulty + economy
You draw	4 + half the difficulty
You lose	1
You abandon	0
You win because your opponent abandons	6
You win without cracking it: your opponent's time bank runs out	10 + difficulty

Difficulty and economy

Rules	Codes	Diff.
Numbers · 3 positions · no repeats	720	6
Numbers · 3 positions · with repeats	1,000	6
Numbers · 4 positions · no repeats	5,040	7
Numbers · 4 positions · with repeats	10,000	8
Numbers · 5 positions · no repeats	30,240	9
Numbers · 5 positions · with repeats	100,000	10
Numbers · 6 positions · no repeats	151,200	10
Numbers · 6 positions · with repeats	1,000,000	12
6 colors · 4 positions · with repeats	1,296	6
8 colors · 6 positions · with repeats	262,144	11



economy by the try on which you crack the code

Economy rewards cracking the code early and runs out on the seventh try.

Difficulty gets 2 more points with limited tries and 2 with a per-turn clock or a time bank; correspondence adds nothing. The cap is 14.

What to take from it

- **In short codes economy pays; in long ones, the rules do.** The longer the code, the more tries it takes, and from the seventh on economy is worth zero: from there everything you earn is paid by the rules.
- **Limited tries and a clock add 2 points each**, however many tries you take.
- **Do not abandon.** A loss is worth 1 point; abandoning, zero. And if it is your opponent who abandons, you get 6: less than any win played to the end.
- **A draw is worth fighting for.** It is 4 points plus half the difficulty: with the last try, play to hit.
- **The arena, the code of the day, the puzzles and practice give no points.** They are where you train.

Three badges you can go after

Badge	How to earn it
Solved in 4	Crack a code in four tries or fewer. With four digits and no repeats, the engine manages it 12.7% of the time; with three, 18%. It takes good play and some luck.
Expert rules	Win by cracking a code with difficulty 12 or more: six digits with repeats; five with repeats or six without, with a clock or with limited tries; or four with repeats, with both.
Five seconds left	Crack the code with five seconds or less on your clock. You need to play with a clock; winning because your opponent runs out of time does not count.

8 Your own code

In a duel you also defend a code. There is no impossible code, but there are codes that give information away.

- **Nothing recognizable.** Runs (1234, 9876), years, dates, the same digit repeated. That is the first thing a person tries.
- **Do not avoid the digits of the openings.** Against someone who opens with 0123, a code with none of those four digits leaves them 360 candidates; one with a single one of them, out of place, leaves 1,440: 4 times as many.
- **With repeats allowed, repeat.** Many people are slow to test quantities: it is the typical mistake from chapter 5.
- **When in doubt, go random.** The “Suggest a random code” button draws one, and a random code has no pattern to exploit.

Against someone who plays well, hardly any code is better than another: what gets punished is the pattern. That is why random is never a bad choice.

9 The other modes, and what they train

The duel is not the only way to play, and each mode trains a different part of the same skill.

Mode	What it is	What it trains
Code of the day	One code a day, the same for everyone: 4 digits without repeats, 8 tries and a single game. The standings rank by tries and then by time.	The second move. The opening does not matter — they are all worth the same —; what separates cracking it in four from cracking it in six is the table in chapter 3. Take your time: time only breaks ties.
Puzzles	72 puzzles in three levels. You get some clues already played and have to deduce the only code that fits. With an account, what you solve follows you to any device.	Pure filtering, with no clock and no opponent: step 3 of the method, until only one is left.
The arena	3 to 8 players against the same code, drawn by the server. No turns, and limited tries. Of the others you only see how many tries they have used and their best number of fijas.	Economy: whoever cracks it in the fewest tries wins. Someone else being on three fijas says nothing about the code: do not change your plan by watching the standings.
Practice	Alone or against the computer, offline and without an account.	All of the above, at no cost. The expert computer plays with this same method and explains every move: read it.

The notebook

A game can be created **with a notebook**: a grid to mark every symbol as ruled out or confirmed, and a warning when the try you are writing **contradicts your own clues**. It is chosen when the game is created, so both players play by the same rules. The marks are yours and never leave the device.

The warning does not say what the code is or how many are left: only that this try cannot be it. It is the most expensive mistake in this guide — playing a code incompatible with a clue — turned into a signal.

The game analysis

When any game ends, “How it played out” goes back over your tries and grades them. It judges the decision, not the luck: it compares what you played with the best possible try **on the same list of candidates**.

Grade	What it means
Optimal	On average, over every possible answer, your try left as few candidates as the best one available, or at most 10% more.
Solid	Close to the best: it splits well, though not perfectly. It leaves at most twice as many.
Wasted	It leaves more than twice as many as the best try. It nearly always repeats information you already had. A try with no fijas and no picas is not wasted for that reason: the grade looks at what it could teach, not at what came out. Tap the grade to see how many it left and which try would have left fewer.
The decisive move	The one after which only one compatible code was left: the game was won there and the rest was writing it down.

Reading the analysis after a loss teaches more than after a win: it points to the exact turn where you stopped splitting and started guessing. It is computed in your browser, from your tries and their clues alone.

10 Score sheet

For printing. One row per try; in the last column, what you deduce or how many candidates you think are left.

Turn	Try	F	P	F+P	Deduction / candidates
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					

In one line. Treat every clue as a condition the code has to meet, rule out whatever does not meet it and choose the next move by what it can teach you, not by how close it looks.